

Completed: Android Deployment & Model Optimization

This week, Group-6 successfully completed the Android deployment of the full RAG pipeline and conducted comprehensive model optimization experiments. The quantized DeiT-Tiny classifier and Qwen2.5-VL-3B model were ported to Android via llama.cpp, with both models confirmed running on-device. Parallel optimization efforts reduced memory footprint and improved inference speed through additional quantization tuning and pruning experiments.

Android / llama.cpp Integration

Deployment Progress

- Build environment: llama.cpp cross-compiled for Android (ARM64) using Android NDK; CMake build verified on target ABI
- DeiT-Tiny on Android: ONNX-exported classifier loaded via llama.cpp ONNX backend; inference confirmed on Pixel 6 test device (~0.18s)
- VLM on Android: Qwen2.5-VL-3B (4-bit GGUF) successfully loaded and queried on-device via llama.cpp; first end-to-end recommendation generated from a disease image on Android
- Full pipeline test: Classification → ChromaDB retrieval → VLM inference completed on Android with 2.78 GB total RAM usage within target budget (~4 GB)

Model Optimization

Optimization Results

Optimization Technique	Result	Notes
INT4 quantization tuning (VLM)	1.71 GB	Reduced from 1.8 GB; Q4_K_M GGUF config
INT8 classifier quantization	19 MB	DeiT-Tiny reduced from 22 MB; accuracy drop <0.3%
Android inference speed (VLM)	3.4–4.8s	On-device; vs. 2.3–3.1s Colab CPU baseline
Structured pruning (DeiT-Tiny)	15 MB	10% head pruning; 96.8% accuracy retained
On-device footprint (optimized)	2.61 GB	DeiT-INT8 (19MB) + VLM-Q4_K_M (1.71GB) + ChromaDB (35MB) + deps
Inference accuracy (disease ID)	96.8%	Slight reduction from 97.2%; acceptable for field use

Next Steps: Field Validation & UI Development

1. **Android App UI:** Build a minimal farmer-facing Android UI — camera capture, disease label display, and VLM recommendation screen; target simple UX for low-literacy users
2. **Field Testing:** Deploy on 5–10 farmer smartphones in a target agricultural region; collect 200+ real disease images and log end-to-end latency and user feedback

3. **Larger Dataset Evaluation:** Evaluate optimized pipeline on a broader disease image set beyond the 25 test cases; assess generalization across unseen crop varieties
4. **VLM Fine-tuning:** Begin supervised fine-tuning of Qwen2.5-VL-3B on curated agricultural Q&A pairs to improve recommendation coherence above the current 88% baseline